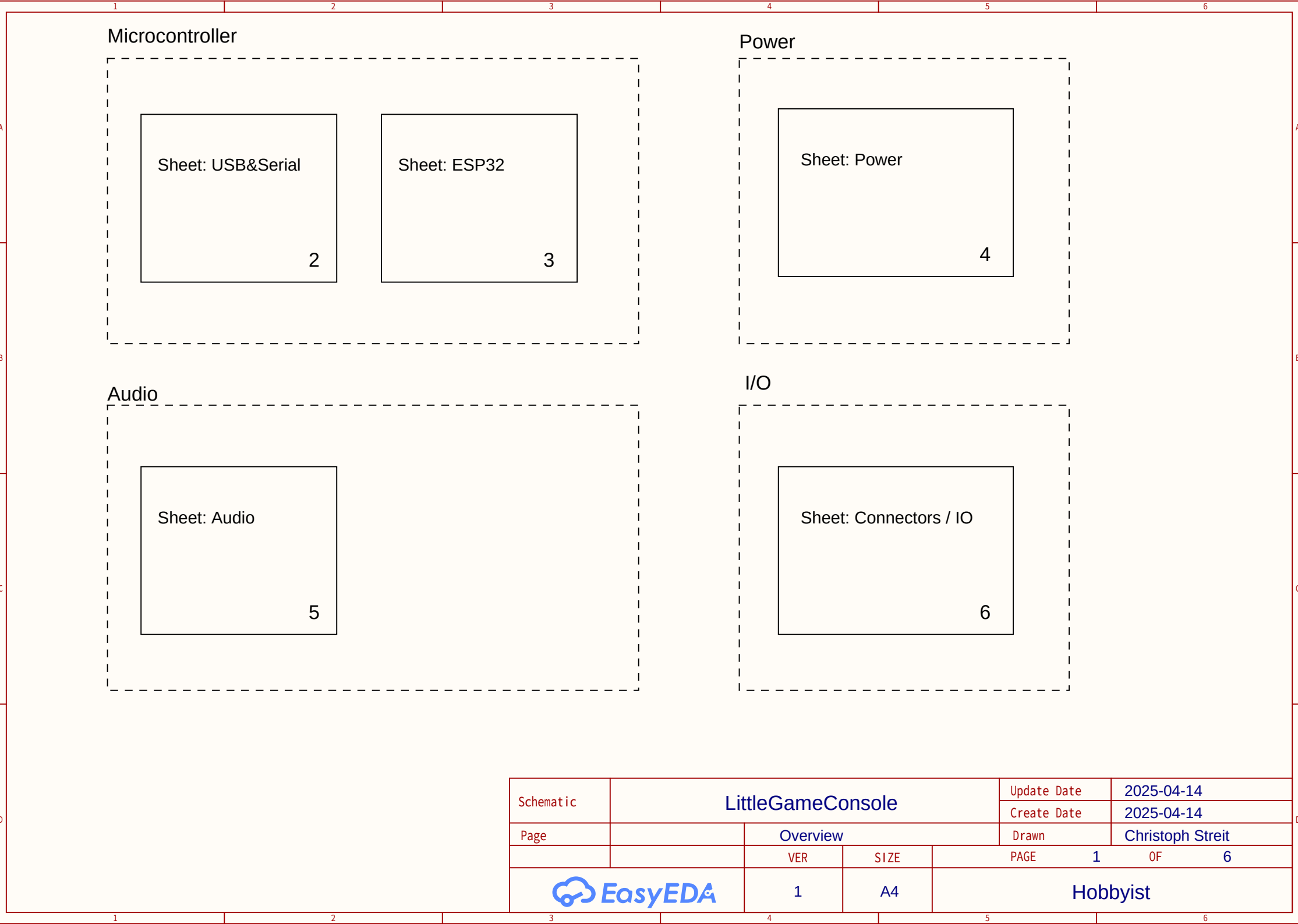
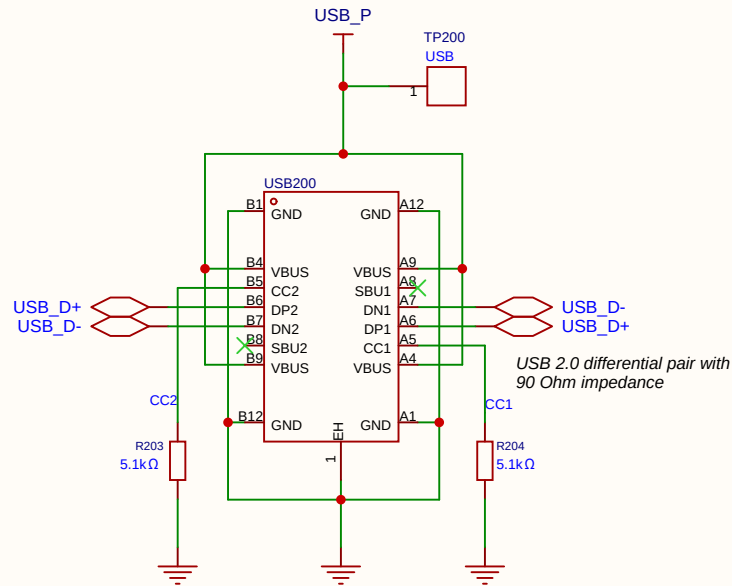


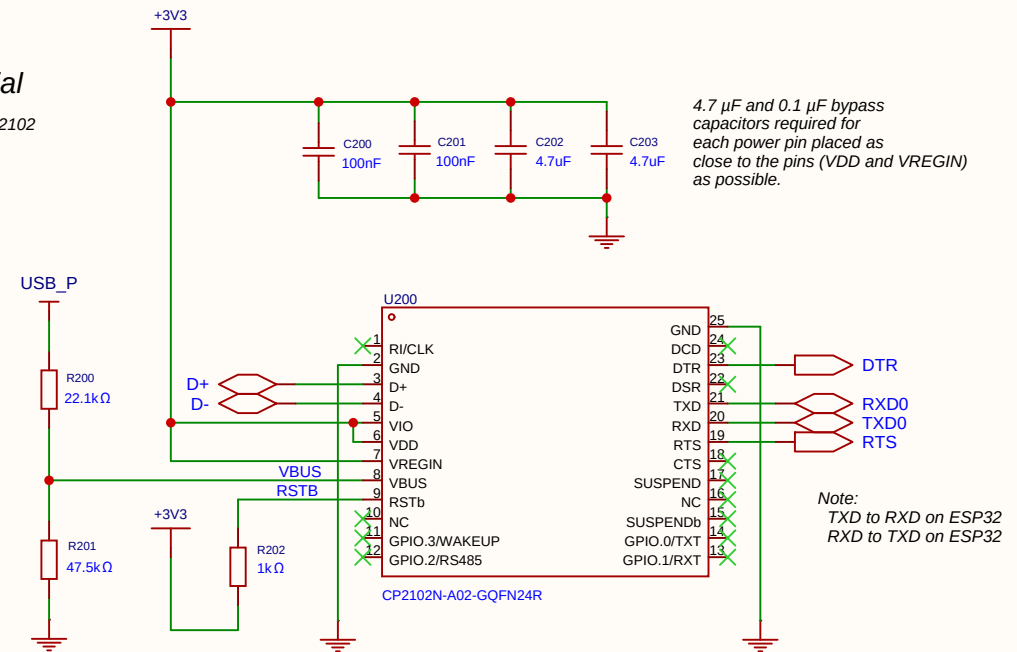


USB



USB to Serial

Bus-powered configuration of the CP2102

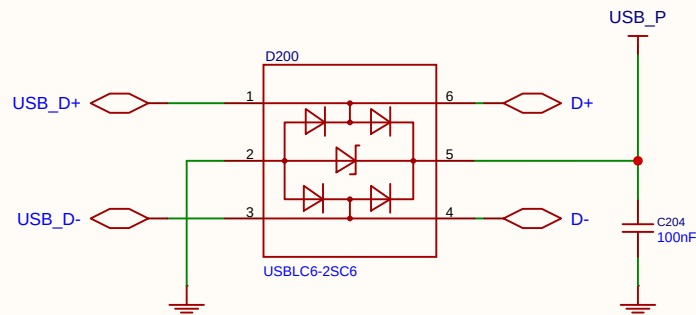


4.7 μ F and 0.1 μ F bypass capacitors required for each power pin placed as close to the pins (VDD and VREGIN) as possible.

Note:
TXD to RXD on ESP32
RXD to TXD on ESP32

Both SUSPEND and SUSPENDb temporarily float high during a CP2102N reset. If this behavior is undesirable, a strong pull-down (10 k Ω) can be used to ensure SUSPENDb remains low during reset.
--> not installed

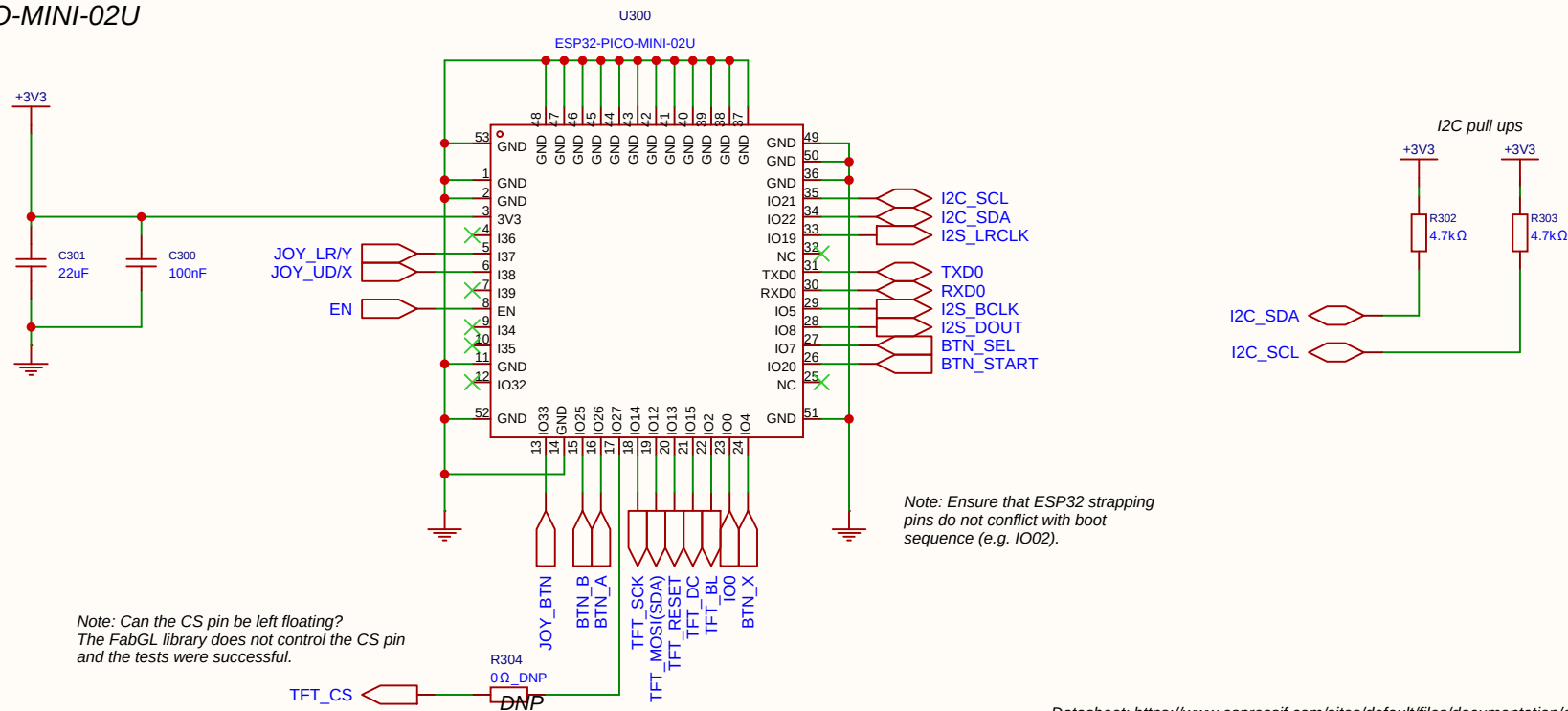
ESD Protection



Datasheet: <https://www.silabs.com/documents/public/data-sheets/cp2102n-datasheet.pdf>

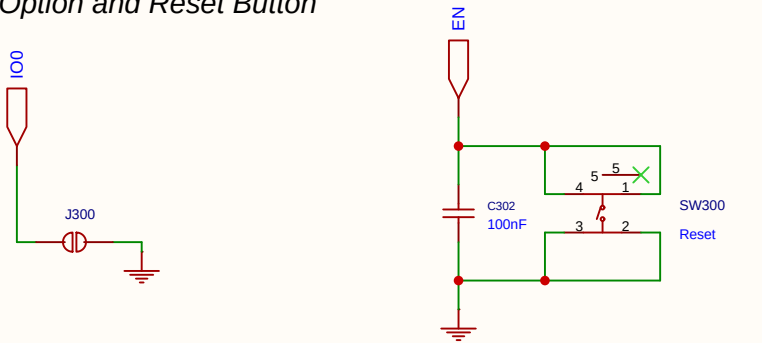
Schematic	LittleGameConsole			Update Date	2025-04-20
				Create Date	2025-04-20
Page	USB+Serial			Drawn	Christoph Streit
		VER	SIZE	PAGE	2 OF 6
EasyEDA		1	A4	Hobbyist	

ESP32-PICO-MINI-02U



Datasheet: https://www.espressif.com/sites/default/files/documentation/esp32-pico-mini-02_datasheet_en.pdf

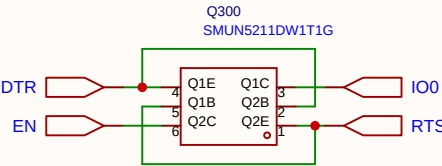
Boot Option and Reset Button



Boot Option to put the board in flashing mode, e.g. short the solder jumper with tweezers.

Press reset button to reset (restart) the board.

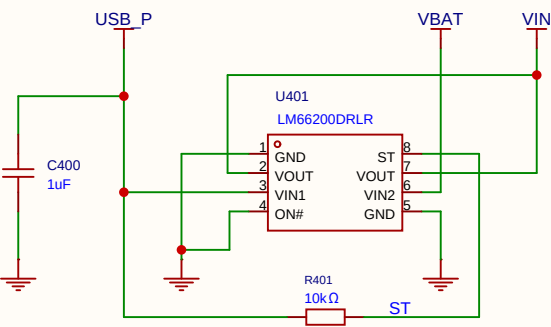
Autoreset



RC delay circuit ensures power supply is stable during power-up

Schematic	LittleGameConsole			Update Date	2025-05-10
				Create Date	2025-05-10
Page	ESP32			Drawn	Christoph Streit
		VER	SIZE	PAGE	3 OF 6
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Automatic Switchover

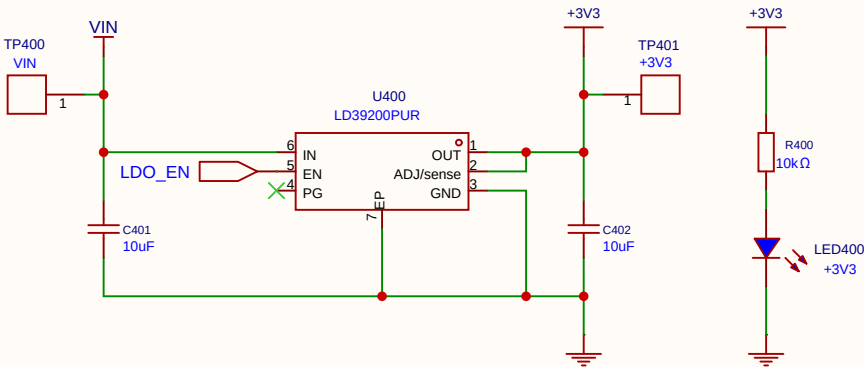


Option: Attach ON# to 1M resistor and then to GND (see Adafruit schematics) in order to turn the device on/off by the CPU

Datasheet: <https://www.ti.com/lit/ds/symlink/lm66200.pdf>

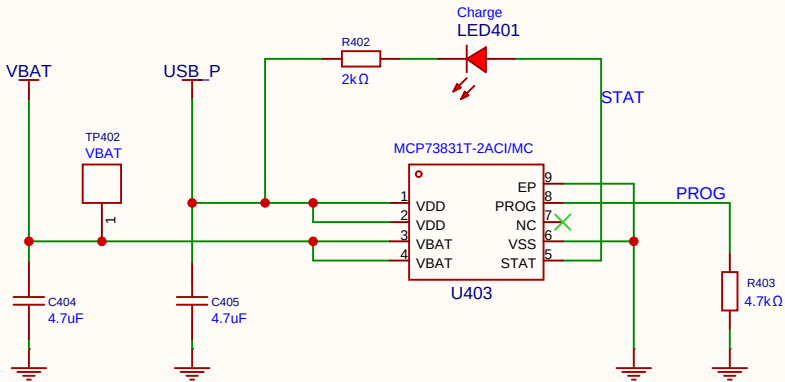
3.3V LDO

3.3V LDO to power the ESP, the CP2102 bridge, the display and other devices



Datasheet: <https://www.st.com/resource/en/datasheet/ld39200.pdf>

Battery Charger

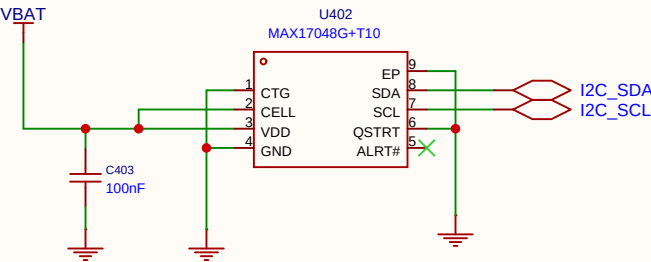


$I_{reg} = 1000V/R_{prog}$
500mA = 1000v/2k (max) -> hot
100mA = 1000v/10k

<https://ww1.microchip.com/downloads/en/DeviceDoc/MCP73831-Family-Data-Sheet-DS20001984H.pdf>

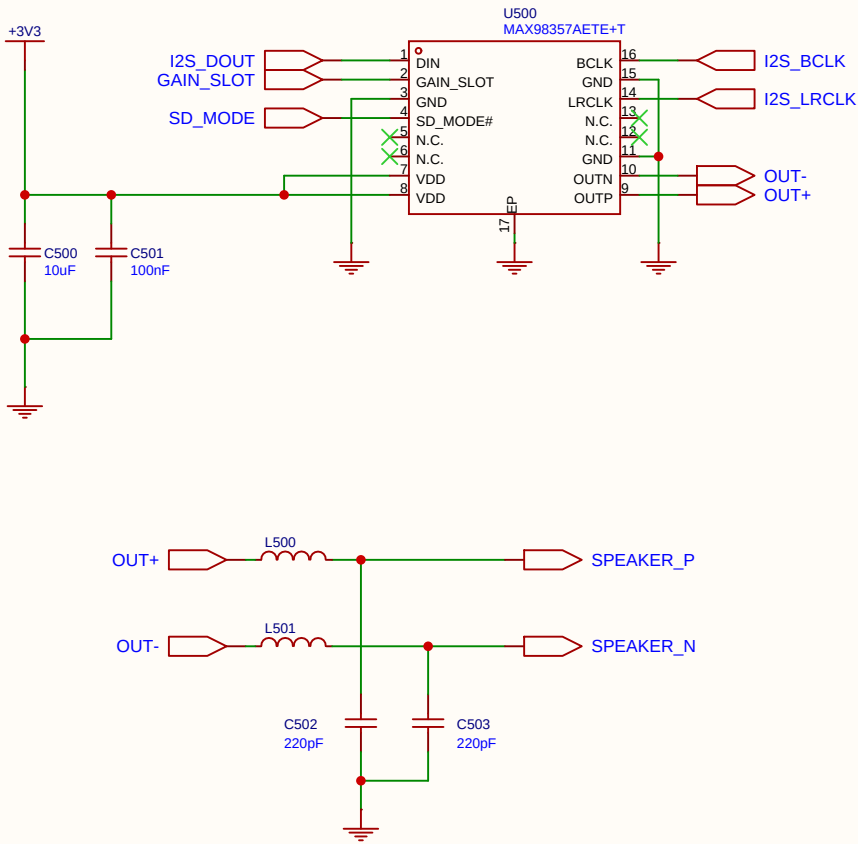
Battery Gauge

Layout hint: VBAT/GND with Kelvin (4-wire) setup

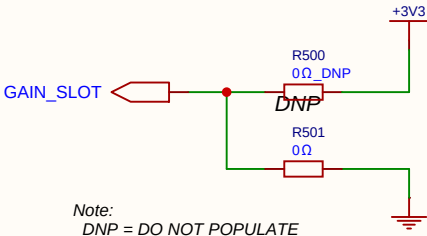


Schematic	LittleGameConsole			Update Date	2025-05-11
				Create Date	2025-05-11
Page		Power		Drawn	Christoph Streit
		VER	SIZE	PAGE	4 OF 6
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Amplifier



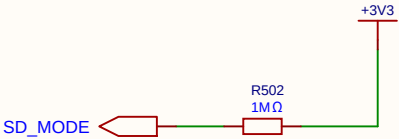
Gain Selection	
GAIN_SLOT	GAIN
Connect to GND through 100kΩ resistor	+15 dB
Connect to GND	+12 dB
Unconnected	+9 dB
Connect to VDD	+6 dB
Connect to VDD through 100kΩ resistor	+3 dB



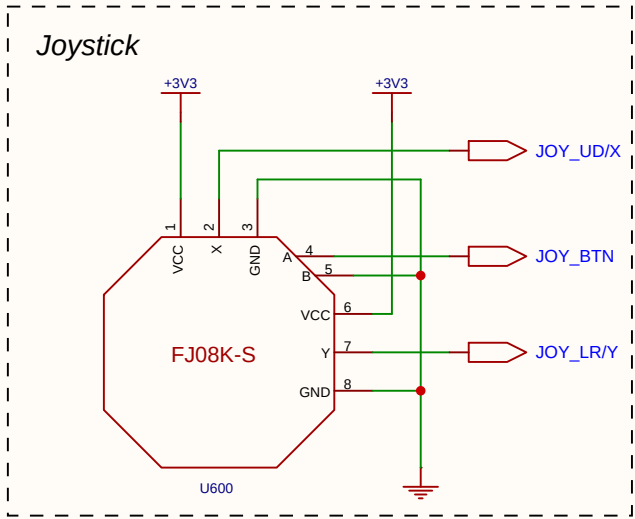
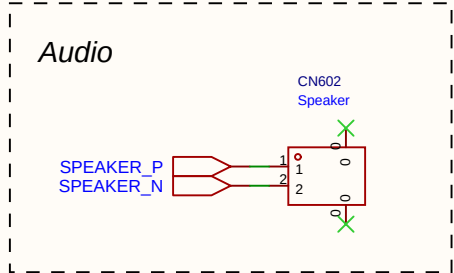
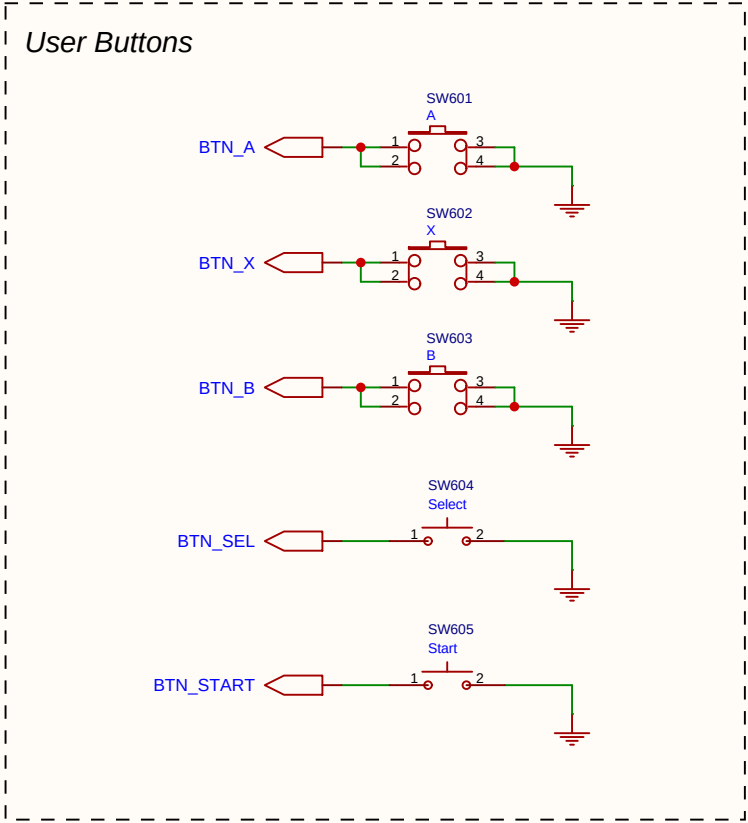
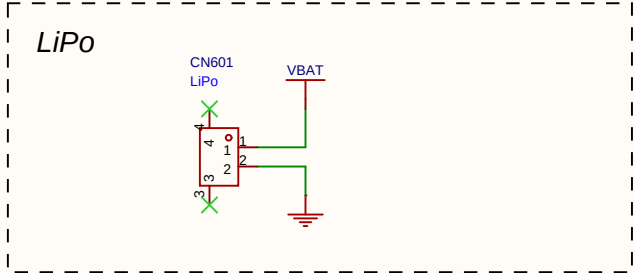
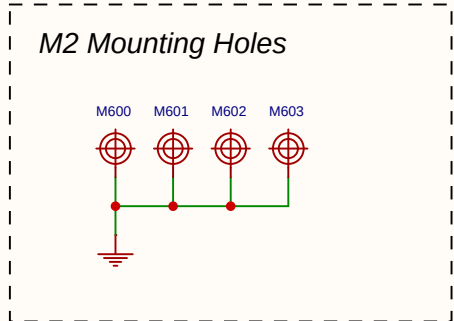
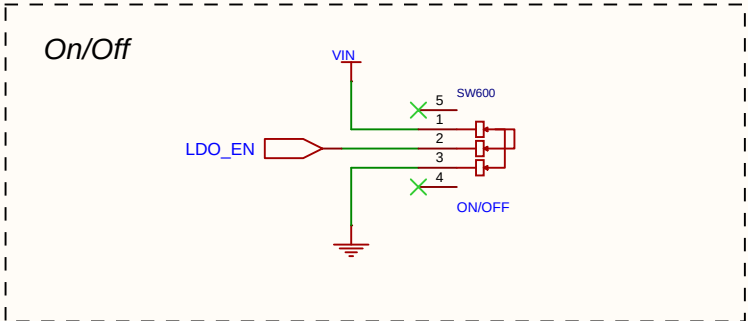
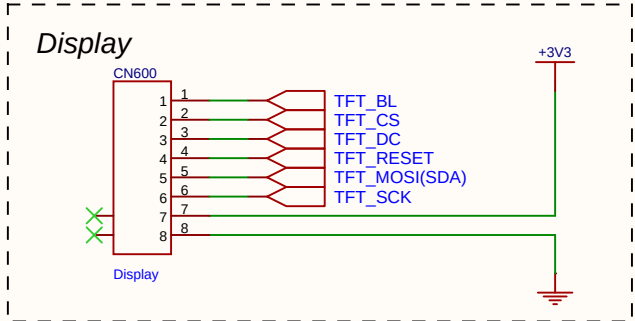
Note:
DNP = DO NOT POPULATE

Channel Selection (SD_MODE)

LEFT = VDD (0 Ω)
RIGHT = 370K
= 94*5-100 (RSMALL (kΩ) = 94.0 x VDDIO - 100)
LEFT/2 + RIGHT/2 = 1.011M
= 222.2*5-100 (RLARGE (kΩ) = 222.2 x VDDIO - 100)



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Schematic	LittleGameConsole			Update Date	2025-05-10
				Create Date	2025-05-10
Page		I/O		Drawn	Christoph Streit
		VER	SIZE	PAGE	6 OF 6
EasyEDA		1	A4	Hobbyist	